

Pharmacogenetic Testing and Therapeutic Drug Monitoring Of Sertraline at a Residential Treatment Center for Children and Adolescents: A Pilot Study.

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Background: Sertraline is commonly prescribed to children for the treatment of anxiety and major depressive disorder and is metabolized in part by CYP2C19. While dosing recommendations based on CYP2C19 genotype exist, there is sparse data in children on the relationship between sertraline concentrations and CYP2C19 genotype. Additionally, although rarely utilized in the United States, therapeutic drug monitoring can also help to guide dosing. The primary objective of this pilot study was to compare sertraline concentrations with CYP2C19 genotype. Secondary objectives included exploring the feasibility of using pharmacogenetic testing and therapeutic drug monitoring in a residential treatment center for children and adolescents. **Methods:** This study was a prospective, open-label study of children prescribed sertraline being treated at a residential treatment center for children and adolescents. Individuals were included if they were < 18 years of age, taking sertraline for at least 2 weeks allowing them to reach steady-state concentrations, being treated through the residential treatment program, and able to understand and speak English. **Results:** A total of 20 participants (80% female) completed all study procedures, including pharmacogenetic testing and therapeutic drug monitoring, with an average age of 15.4 years (range: 9-17 years). Forty percent (n=8) of participants had a diagnosis of Generalized Anxiety Disorder, while 30% (n=6) had a diagnosis of Major Depressive Disorder. Overall, average sertraline and desmethylsertraline concentrations were 21.1 ng/ml (range: 1-78 ng/ml) and 52.4 ng/ml (range: 1-258 n/ml). Based on CYP2C19 genotypes, 60% (n=12) were normal metabolizers, 10% (n=2) were intermediate metabolizers, and 30% (n=6) were rapid metabolizers. Daily sertraline dose (mg/day) accounted for a significant amount of the observed variability in sertraline ($p<0.0001$; $r^2=0.62$) and desmethylsertraline concentrations ($p<0.001$; $r^2=0.45$). When comparing weight-based dosing by sertraline and desmethylsertraline concentrations, sertraline daily dose by weight (mg/kg/day) also accounted for a significant amount of the observed variability in sertraline ($p<0.0001$; $r^2=0.60$) and desmethylsertraline ($p<0.0001$; $r^2=0.59$) concentrations. Average daily and weight-based doses for CYP2C19 intermediate, normal, and rapid metabolizers were 75 mg/day, 87.5 mg/day, and 79.2 mg/day and 1.5 mg/kg/day, 1.3 mg/kg/day, and 1.1 mg/kg/day, though these were not significantly different. **Conclusion:** This small, pilot study showed sertraline dose to be significantly associated with sertraline and desmethylsertraline concentrations. No differences were noted between CYP2C19 metabolizer groups, likely due to the limited sample size. These results also suggest that ordering pharmacogenetic testing and therapeutic drug monitoring in the setting of a child and adolescent residential treatment center is feasible.

Sertraline in children and adolescents with major depressive disorder.

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To explore time to first response and time to first persistent response of sertraline versus placebo and compare these parameters between children (6-11 years old, n = 177) and adolescents (12-17 years old, n = 199) with major depressive disorder. A 10-week placebo-controlled treatment was followed by a 24-week open-label sertraline treatment. The double-blind studies were not powered to detect efficacy differences between age groups. A post hoc analysis explored time to first response and first persistent response using the Children's Depression Rating Scale-Revised and Clinical Global Impressions-Improvement predefined criteria. There were no statistically significant differences in time to first response or first persistent response between sertraline and placebo in children, except for time to

first response on Clinical Global Impressions-Improvement. Sertraline had a significantly faster time to first persistent response in adolescents compared to placebo. Within treatment groups, children had a significantly faster time to first response than adolescents, whether treated with placebo or sertraline, but not on time to first persistent response. Both age groups showed similar improvement over 34 weeks of treatment. In the double-blind studies, children and adolescents had different patterns of response with sertraline vs. placebo.

[Sertraline in child and adolescent psychiatry].

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The aim of the study was to research the efficiency of sertraline (zoloft) in depressions, anxious states and obsessive-compulsive disorders. Diagnosis of the mental disorders was carried out according to ICD-10. 72 children (59 boys, 13 girls) aged 6-18 years were treated. There were 32 inpatients and 40 outpatients. Therapy with sertraline was performed during 8 weeks with a gradual increase (titration) and individual selection of the dose from 12.5 to 100 mg/day. During the therapy clinical observation was combined with the patients' examination using Hamilton Depression Scale and Hamilton Anxiety Scale (HAM-D and HAM-A), and a Clinical Global Impression Scale (CGI). It was established that sertraline was very effective and safe drug in children (it has no influence on cognitive functions, has neither myorelaxing or sedative effects). Activity of this drug is characterized by quick manifestation of thymoanaleptic and anxiolytic effects. It mild depressive states 50 mg/day is a significant dose; in more severe depressions and obsessive-compulsive disorders the dose in juveniles was to 100 mg, the duration of the therapy was more than 2 months.

Asynchronicity of Organic and Psychiatric Symptoms in a Case of Sertraline Intoxication.

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Sertraline is a selective serotonin reuptake inhibitor used as an antidepressant and antipanic agent in children and adolescents. Sertraline is well tolerated and its safety profile in overdose is favorable. However, sertraline overdose may cause a toxic hyperserotonergic state known as serotonin syndrome (SS). Serotonin syndrome may be misdiagnosed in children because it has been reported mostly in adults. In the present case report, we describe a 16-year-old female patient who ingested 2000 mg of sertraline to attempt suicide. The patient showed symptoms and signs suggestive of SS, characterized by an asynchronicity between organic and psychiatric symptoms. In addition, the patient showed a variability of psychiatric symptoms through time. Thirteen hours after sertraline overdose, she was poorly cooperative with poverty of speech and marked emotional tension, but she was oriented in space and time and was able to remember what happened to her and to reconstruct the dynamics of the fact. Twenty-four hours after sertraline overdose, the patient developed a delusion of reference associated with intense anxiety and depressed mood. In the present case report, we discuss the pathophysiologic mechanisms of the observed clinical manifestations and propose an observation period for sertraline overdose in children and adolescents that is sufficiently long (at least 72 hours) even in the absence of unstable vital signs.

Molecular Biomarkers Predictive of Sertraline Treatment Response in Young Children With Autism Spectrum Disorder.

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Sertraline is one among several selective serotonin reuptake inhibitors (SSRIs) that exhibited improvement of language development in Autism Spectrum Disorder (ASD); however, the molecular mechanism has not been elucidated. A double blind, randomized, 6-month, placebo-controlled, clinical trial of low-dose sertraline in children ages (3-6 years) with ASD was conducted at the UC Davis MIND Institute. It aimed at evaluating the efficacy and benefit with respect to early expressive language development and global clinical improvement. This study aimed to identify molecular biomarkers that might be key players in the serotonin pathway and might be predictive of a clinical response to sertraline. Fifty eight subjects with the diagnosis of ASD were randomized to sertraline or placebo. Eight subjects from the sertraline arm and five from the placebo arm discontinued from the study. Furthermore, four subjects did not have a successful blood draw. Hence, genotypes for 41 subjects (20 on placebo and 21 on sertraline) were determined for several genes involved in the serotonin pathway including the serotonin transporter-linked polymorphic region (5-HTTLPR), the tryptophan hydroxylase 2 (TPH2), and the Brain-Derived Neurotrophic Factor (BDNF). In addition, plasma levels of BDNF, Matrix metalloproteinase 9 (MMP-9) and a selected panel of cytokines were determined at baseline and post-treatment. Intent-to-treat analysis revealed several primary significant correlations between molecular changes and the Mullen Scales of Early Learning (MSEL) and Clinical Global Impression Scale - Improvement (CGI-I) of treatment and control groups but they were not significant after adjustment for multiple testing. Thus, sertraline showed no benefit for treatment of young children with ASD in language development or changes in molecular markers in this study. These results indicate that sertraline may not be beneficial for the treatment of children with ASD; however, further investigation of larger groups as well as longer term follow-up studies are warranted.

Exploring the immunomodulatory effects of sertraline: Cytokine modulation and signaling pathway dynamics.

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This study explores the nuanced immunomodulatory effects of sertraline, which is widely used in the treatment of major depression, obsessive-compulsive disorder, and anxiety in adults and children. Recent investigations have emphasized the intricate interplay between depression and the body's inflammatory response. This has sparked an exploration into the impact of sertraline on the immune system, an area that still awaits comprehensive exploration. Our research methodically examines the influence of sertraline on the levels of cytokines (TNF- α , IL-6, IL-12p40, GM-CSF) in the macrophage cell line J774.2. This analysis is conducted under conditions with and without the lipopolysaccharide (LPS) danger signal. To enhance specificity, sertraline's effects are juxtaposed with those of salicylic acid, a known anti-inflammatory agent. Furthermore, a comprehensive exploration of sertraline's impact on the intracellular signaling pathways regulated phosphoinositide-3-kinase (PI3K) and the p38 pathway is the third major signaling cassettes of the mitogen-activated protein kinase (MAPK) signaling is presented. The outcomes of our study unveil distinctive patterns in sertraline's modulation of cytokine levels within macrophage cells. Under the influence of the LPS danger signal, sertraline exhibits immunostimulatory characteristics, contrasting with its ability to suppress GM-CSF cytokine levels, even in the presence of LPS. Notably, the p38 pathway portrays a pro-inflammatory role for sertraline, while inhibiting the PI3K

signaling pathway highlights its anti-inflammatory attributes. These findings contribute novel insights into the intricate interplay between sertraline and the immune system.

Sertraline may improve language developmental trajectory in young children with fragile x syndrome: a retrospective chart review.

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Young children with fragile X syndrome (FXS) often experience anxiety, irritability, and hyperactivity related to sensory hyperarousal. However, there are no medication recommendations with documented efficacy for children under 5 years old of age with FXS. We examined data through a chart review for 45 children with FXS, 12-50 months old, using the Mullen Scales of Early Learning (MSEL) for baseline and longitudinal assessments. All children had clinical level of anxiety, language delays based on MSEL scores, and similar early learning composite (ELC) scores at their first visit to our clinic. Incidence of autism spectrum disorder (ASD) was similar in both groups. There were 11 children who were treated with sertraline, and these patients were retrospectively compared to 34 children who were not treated with sertraline by chart review. The baseline assessments were done at ages ranging from 18 to 44 months (mean 26.9, SD 7.99) and from 12 to 50 months (mean 29.94, SD 8.64) for treated and not treated groups, respectively. Mean rate of improvement in both expressive and receptive language development was significantly higher in the group who was treated with sertraline ($P < 0.0001$ and $P = 0.0071$, resp.). This data supports the need for a controlled trial of sertraline treatment in young children with FXS.

Sertraline-Induced Rhabdomyolysis, Trismus, and Cardiac Arrest in a Child.

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Selective serotonin reuptake inhibitors are a commonly used and often effective class of medications in the treatment of mood disorders such as anxiety and depression. Sertraline (1S,4S-N-methyl-4-[3,4-dichlorophenyl]-1,2,3,4-tetrahydro-1-naphthylamine [Zoloft; Pfizer, New York City, NY]) is a frequently used selective serotonin reuptake inhibitor that has shown efficacy in children, adolescents, and adults. We report the case of a 13-year-old boy with sertraline-induced rhabdomyolysis and renal failure, trismus, and cardiopulmonary arrest. Pharmacogenetic testing later revealed our patient had serotonin transporter polymorphisms and enzymatic alterations that put him at risk for increased levels of sertraline and greater likelihood for untoward side effects.

A Randomized Controlled Trial of Sertraline in Young Children With Autism Spectrum Disorder.

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Objective: Selective serotonin reuptake inhibitors like sertraline have been shown in observational studies and anecdotal reports to improve language development in young children with fragile X syndrome (FXS). A previous controlled trial of sertraline in young children with FXS found significant

improvement in expressive language development as measured by the Mullen Scales of Early Learning (MSEL) among those with comorbid autism spectrum disorder (ASD) in post hoc analysis, prompting the authors to probe whether sertraline is also indicated in nonsyndromic ASD. Methods: The authors evaluated the efficacy of 6 months of treatment with low-dose sertraline in a randomized, double-blind, placebo-controlled trial in 58 children with ASD aged 24 to 72 months. Results: 179 subjects were screened for eligibility, and 58 were randomized to sertraline (32) or placebo (26). Eight subjects from the sertraline arm and five from the placebo arm discontinued. Intent-to-treat analysis showed no significant difference from placebo on the primary outcomes (MSEL expressive language raw score and age equivalent combined score) or secondary outcomes. Sertraline was well tolerated, with no difference in side effects between sertraline and placebo groups. No serious adverse events possibly related to study treatment occurred. Conclusion: This randomized controlled trial of sertraline treatment showed no benefit with respect to primary or secondary outcome measures. For the 6-month period, treatment in young children with ASD appears safe, although the long-term side effects of low-dose sertraline in early childhood are unknown. Clinical Trial Registration: www.ClinicalTrials.gov, identifier NCT02385799.

A case of isolated pachygyria with unusual clinical onset in the neonatal period.

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Pachygyria is a malformation of the cerebral cortex characterized by an insufficient development of the gyri and cerebral sulci within the anomalies of neuronal migration. Clinically, can cause seizures, mental retardation and motor deficits. We report a rare clinical case of pachygyria in a newborn with persistent hypotonia and sub-clinical seizures in which early diagnosis was possible thanks to cranial magnetic resonance imaging.
